

Ultimate Nutrition

MELATONIN

Being too tired, either due to hard work, lack of exercise or too intense an exercise regime, can throw the body's "clock" into disarray. The result is lack of restful sleep, which can cause a myriad of reactions including restlessness, nervousness, irritation and lost productivity. Melatonin is the natural hormone that the body produces to regulate its internal "clock."

Melatonin is produced by the pineal gland, which is a pea-sized gland at the base of the brain. Even though the exact functions of melatonin are still under study, it is involved in the synchronization of hormone secretion. This natural daily cycle of hormone release in the body is commonly referred to as circadian rhythm. The human body is under the influence of an internal clock, which controls the secretion of various hormones at different times of the day for the body to function properly. The secretion of melatonin is triggered by the dark and is suppressed by natural daylight, therefore controlling periods of sleepiness and wakefulness.

However, melatonin is more than a "sleeping pill". It can also strengthen the immune system. It works in concert with vitamin B6 and zinc, for instance, to reverse the decline of the immune system that comes with advancing age. In addition, melatonin has strong antioxidant activity and, as such, it supports the hormonal and metabolic networks in the body. These networks undergo additional changes due to intense physical exercise, mental and emotional stress, or even poor nutrition. Some evidence suggests that melatonin may, in fact, lighten and enhance one's mood considerably. Melatonin is also reported to have a role in improving women's health problems, such as osteoporosis and premenstrual syndrome.

Ultimate Nutrition's Melatonin is especially designed and manufactured for performance athletes who occasionally may need nature's helping hand to get their rest, and for traveling executives to better control their sleep cycle. Importantly, melatonin restores the natural hormonal and metabolic balance in the body for overall health.

SELECTED REFERENCES

Mallo et al., "Effects of a Four-Day Nocturnal Melatonin Treatment on the 24-Hour Plasma Melatonin, Cortisol, and Prolactin Profiles in Humans," *Acta Endocrinol.*: 119, 474, 1995

Retier et al., "A Review of the Evidence Supporting Melatonin's Role as an Antioxidant," *J. Pineal Res.*: 18, 1, 1995

Reiter et al., "Oxygen Radical Detoxification Processes During Aging: The Functional Importance of Melatonin," Aging Clin. Exp. Res.: 7, 340, 1995

Lewis, A. & Clouatre, D., Melatonin and the Biological Clock, New Canaan, Connecticut: Keats Publishing, 1996

Goldberg, B., Alternative Medicine Guide to Women's Health, Tiburon, California: Future Medicine Publishing, 1998

FREQUENTLY ASKED QUESTIONS

1. Is Melatonin the Same as Melanin?

No; melatonin and melanin are two distinct compounds, but are usually confused with each other. Melanin is the compound responsible for skin pigmentation whereas melatonin is the hormone that regulates circadian rhythm in the body. Therefore, it controls the sleep and waking cycles.

2. What is the Recommended daily Dose for Melatonin?

Daily dosages ranging from 3 mg per day to 10 mg per day have been recommended. That may be too high, since at least one study has shown that intake of 8 mg melatonin per day may significantly alter circadian rhythm as changes in the pattern of hormone secretion alter. Melatonin should certainly be taken in microgram amounts. The dosage taken should be tailored to individual needs and tolerance levels. Self-initiated regimen of melatonin intake should start with the lowest dose, which could be escalated as and when needed, being aware of one's tolerance levels.

3. When is Melatonin Best Taken?

Melatonin should be taken ONLY in the evening when one does not anticipate driving or working with machinery or running errands. It should be taken a few hours before retiring. It should be emphasized that timing is important. Taking melatonin at the wrong time --i.e., time opposing the normal daily cycle-- can weaken immunity and is likely to show other unhealthful effects. For optimal dosage and the correct time window, the primary care physician should be consulted prior to intake of melatonin.

4. Under What Conditions is Melatonin NOT Recommended?

Melatonin intake is not recommended in cases where antidepressants and MAO (monoamine oxidase) inhibitors have been prescribed. In addition, melatonin should not be taken in cases of autoimmune diseases due to the effect of melatonin on the levels of hormones produced by the adrenal gland. Adrenal cortical hormones are physiological inhibitors of autoimmunity and, hence, the use of melatonin in autoimmune conditions is not recommended. Is Melatonin safe for Pregnant or Lactating Women? Melatonin regulates hormonal level throughout the body. Consequently, melatonin is not deemed appropriate for pregnant and/or lactating women.

5. What Are the Side Effects of Melatonin?

Melatonin is considered safe and without side effects for most people if used intermittently and in proper doses as an aid to synchronize the body's internal clock, such as in cases of jet lag, shift work, etc.